

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

Computer Graphics: Lecture 1

Welcome to Real-time 3D!

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Welcome!

You have done some cool things as a computer science major:

- You have learned and implemented a wide range of algorithms...
- Data structures too...
- You have developed user-facing applications (GUIs)

You have learned many languages, and built many projects, to do all sorts of cool things.

However, there is a set of capabilities inside your computer that you have not even *begun* to unlock. The computer is **way** more powerful than you've been able to take advantage of.

The computer's superpower

Almost every modern computer (including phones) contains a powerful **co-processor** called a GPU—a graphics processing unit. This GPU might be on a special expansion card, like this RTX 5090 here:



GPUs are
incredibly
powerful.

They can produce
beautiful 3D scenes
like this one dozens or
hundreds of times per
second

Questions?